

# FILTER DESIGN TEST CASE

Coupled Resonator Filter - Band 20 (800 MHz) for WiFi 6E Coexistence Filter  
Design Validation and Performance Characterization

## DOCUMENT INFORMATION

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|------------------|----------------------------|
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| Author:          | Dr. Ahmed Al-Farsi         |
| Reviewer:        | Dr. Olaf Bergstrom         |
| Design Center:   | Newbury Park Design Center |
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| Status:          | CONDITIONAL PASS           |
| Priority:        | P4 - Low                   |

## 1. OBJECTIVE

This document describes the design, simulation, and characterization of a Coupled Resonator Filter filter targeting Band 20 (800 MHz) for WiFi 6E Coexistence Filter. The filter is designed on AIN on Si substrate with a target center frequency of 4148.5 MHz and bandwidth of 231.1 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the WiFi 6E Coexistence Filter module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

## 2. SCOPE

- Filter Type: Coupled Resonator Filter
- Frequency Band: Band 20 (800 MHz)
- Application: WiFi 6E Coexistence Filter
- Substrate: AIN on Si
- Package: Fan-Out WLP
- Design Tool: Synopsys CustomSim
- Design Center: Newbury Park Design Center

This specification applies to all variants of the Coupled Resonator Filter design targeting Band 20 (800 MHz). Results are used to qualify the design for mass production release.

## 3. DESIGN PARAMETERS

|                          |                |
|--------------------------|----------------|
| Center Frequency:        | 4148.5 MHz     |
| Bandwidth (3 dB):        | 231.1 MHz      |
| Insertion Loss Target:   | $\leq 3.2$ dB  |
| Return Loss Target:      | $\geq 15.0$ dB |
| Out-of-Band Rejection:   | $\geq 45.0$ dB |
| Isolation (Duplexer):    | $\geq 21.0$ dB |
| Q Factor:                | 4210           |
| Temperature Coefficient: | -16.1 ppm/°C   |
| Die Size:                | 2.7 x 0.6 mm   |
| Wafer Size:              | 6-inch         |
| Number of Resonators:    | 3              |
| Electrode Material:      | W              |

Passivation: SiO2

4. TEST PROCEDURE

1. Open Synopsys CustomSim and load the Coupled Resonator Filter template project.
2. Define the substrate stack: AlN on Si with specified layer thicknesses.
3. Set design targets: center freq 4148.5 MHz, BW 231.1 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 3.2 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (Fan-Out WLP).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (6-inch, AlN on Si).
10. Perform on-wafer probing using Rohde & Schwarz FSW Signal Analyzer.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 12446 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 3.2 dB
- Return loss in passband shall be >= 15.0 dB
- Out-of-band rejection at +/- 347 MHz offset >= 45.0 dB
- Group delay variation across passband shall be <= 8 ns
- Temperature drift over -40°C to +85°C shall be <= 8.3 MHz
- Power handling shall be >= +31 dBm at 1 dB compression
- ESD tolerance >= 2000 V (HBM)
- Die yield on qualification lot >= 96%

6. TEST RESULTS

Measurement Date: 2024-03-30  
Devices Tested: 30  
Insertion Loss (meas): 2.94 dB  
Return Loss (meas): 16.8 dB  
Rejection (meas): 48.1 dB  
Isolation (meas): 26.3 dB  
Q Factor (meas): 4210  
Group Delay Variation: 7.8 ns  
Power Handling: +30 dBm  
Die Yield: 88.1%

Result Classification: CONDITIONAL PASS

7. OBSERVATIONS AND FINDINGS

- a) The Coupled Resonator Filter design demonstrated acceptable performance for Band 20 (800 MHz).
- b) Insertion loss is marginally above target; electrode thickness optimization recommended.
- c) Out-of-band rejection exceeded the minimum requirement.
- d) Temperature stability was within specification across full range.
- e) Batch-to-batch reproducibility confirmed over 3 wafer lots.

8. CORRECTIVE ACTIONS

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- 1. Review near-band rejection margin for worst-case temperature.
- 2. Re-run EM simulation with refined mesh for better accuracy.
- 3. Schedule follow-up measurement after design tweak.

9. SIGN-OFF

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|-----------|--------------------|-------|------------|
| Author:   | Dr. Ahmed Al-Farsi | Date: | 2024-03-30 |
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| Approver: | Dr. Karen Mitchell | Date: | 2024-04-12 |

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